

**Owner-approved ExamSIDE question-demand
slice for plane motion, now semantically:
Supported practice**

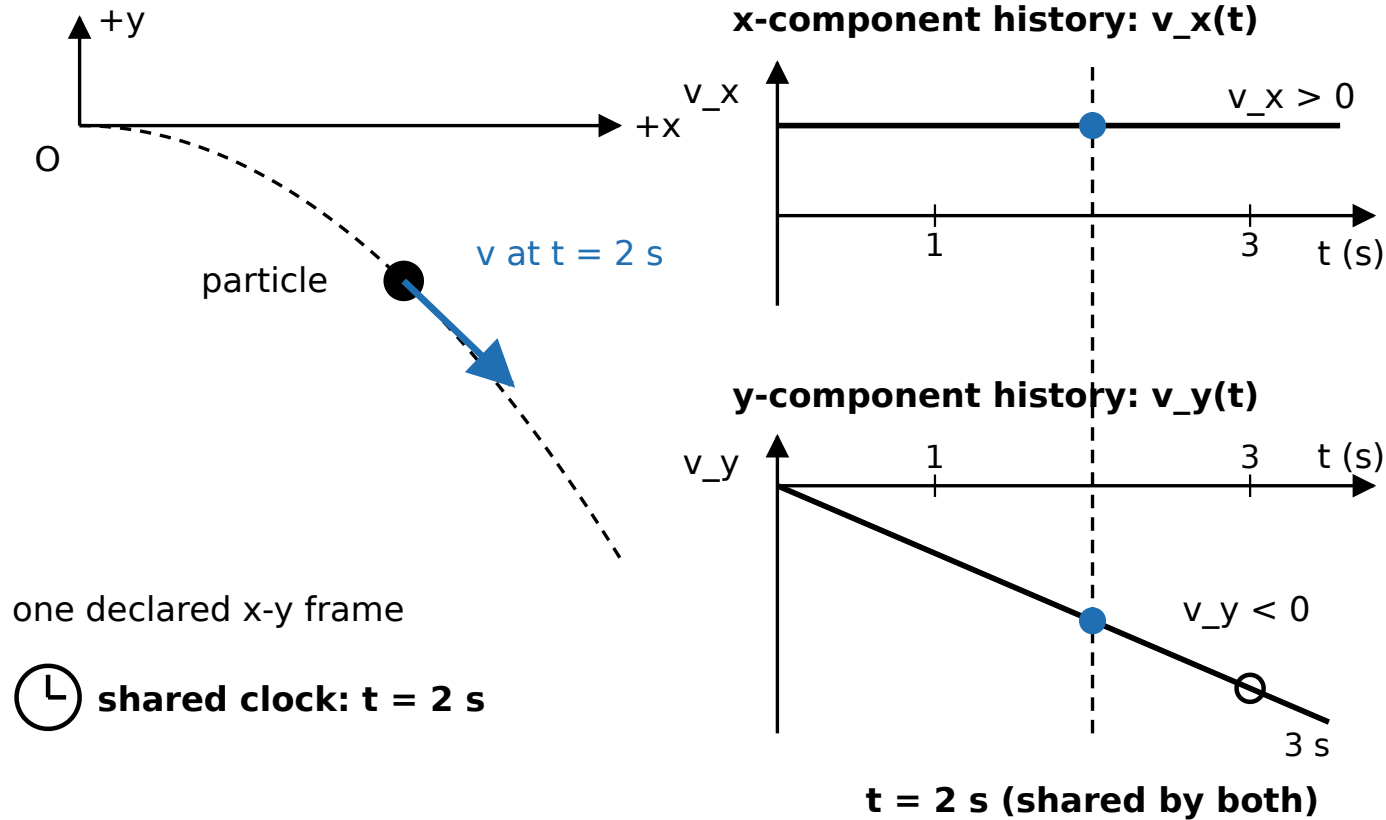
AUTHORED practice

Motion in 2D component and projectile practice

A tracker reports $x(t)=4t$ metres and $y(t)=10-2t^2$ metres for one particle in one fixed frame. Find the particle's position at $t=2$ s, and explain why $x(2$ s) may not be paired with $y(3$ s) to describe one position.

Conditions

- The two functions refer to the same particle and the same time coordinate.
- Use the declared fixed x-y frame.



✓ $(v_x(2\text{ s}), v_y(2\text{ s}))$: simultaneous, one valid plane state

✗ $(v_x(2\text{ s}), v_y(3\text{ s}))$: non-simultaneous, rejected (no single instant)

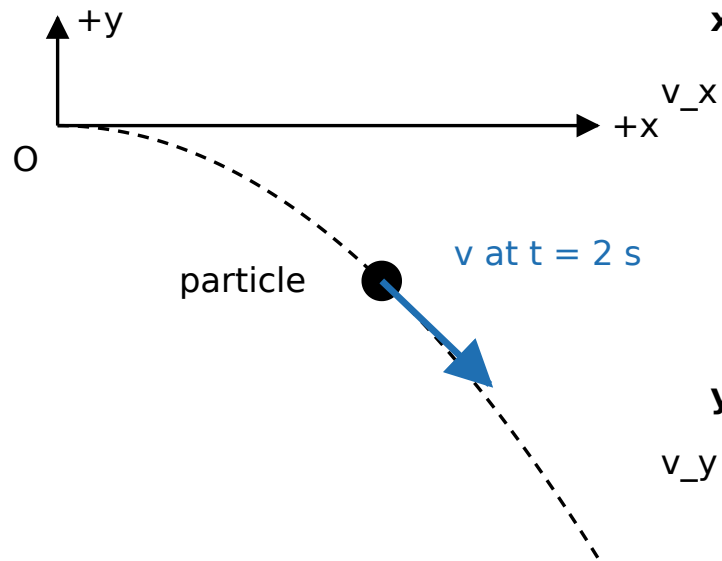
Keep independently evolving x and y descriptions attached to one physical object, one declared frame and one shared clock, so only simultaneous component values are recombined into a plane state.

1. What must be the same when $x(t)$ and $y(t)$ describe one particle at one instant?

Reasoning route and full solution

1. **REPRESENT** Treat $x(t)$ and $y(t)$ as two coordinate projections of the same particle in one fixed frame, tied to one common event time t .
Why valid: A single plane position is defined by simultaneous coordinates of one particle; independent component functions do not create independent clocks.
2. **TRANSFORM** Substitute $t=2$ s into both component functions to obtain $x(2)=8$ m and $y(2)=2$ m.
Why valid: The requested state is the event labelled $t=2$ s, so both component functions must be sampled at that event.
3. **CONNECT** Pair the two simultaneous values as the position $(8,2)$ m and reject $x(2)$ paired with $y(3)$ as a single physical state.
Why valid: Coordinates form one state only when they refer to the same object, frame and instant; changing one timestamp changes the event.
4. **VERIFY** Attach the timestamp to each component value and confirm the two timestamps match before calling the pair a position.

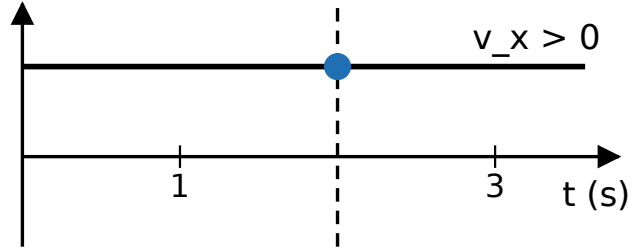
Why valid: Timestamp agreement is an independent event-identity check that does not depend on the arithmetic values themselves.



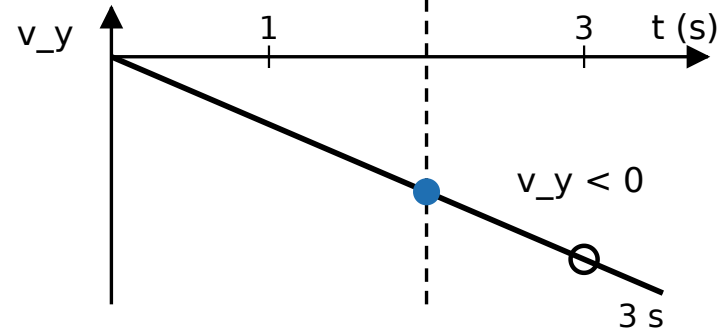
one declared x-y frame

⌚ shared clock: $t = 2 \text{ s}$

x-component history: $v_x(t)$



y-component history: $v_y(t)$



$t = 2 \text{ s}$ (shared by both)

✓ **$(v_x(2 \text{ s}), v_y(2 \text{ s}))$: simultaneous, one valid plane state**

✗ $(v_x(2 \text{ s}), v_y(3 \text{ s}))$: non-simultaneous, rejected (no single instant)

Keep independently evolving x and y descriptions attached to one physical object, one declared frame and one shared clock, so only simultaneous component values are recombined into a plane state.

1. Evaluate both component descriptions at the same requested time.
2. $x(2)=4(2)=8$.
3. $y(2)=10-2(2^2)=2$.
4. A physical plane state is a simultaneous component pair.

Answer

At $t=2$ s, $x=8$ m and $y=2$ m, so the position is (8,2) m in the declared frame. $x(2$ s) and $y(3$ s) cannot form one position because they refer to different instants.

Independent check

The answer must use one t for both components; mixing timestamps produces no real single-time state.

If you went wrong

Your position is (8, -8) m or (12, 2) m: one coordinate was read at $t = 3$ s ($y(3) = -8$ m, $x(3) = 12$ m) while the other used $t = 2$ s.

Repair

Revisit: Recombine only component values that refer to the same frame and the same instant.

What this establishes

Established: a plane position is the pair $x(t)$, $y(t)$ read at one instant in one frame. The transfer task keeps the one frame and one clock but removes the given component functions, so the signed components must be built from words.

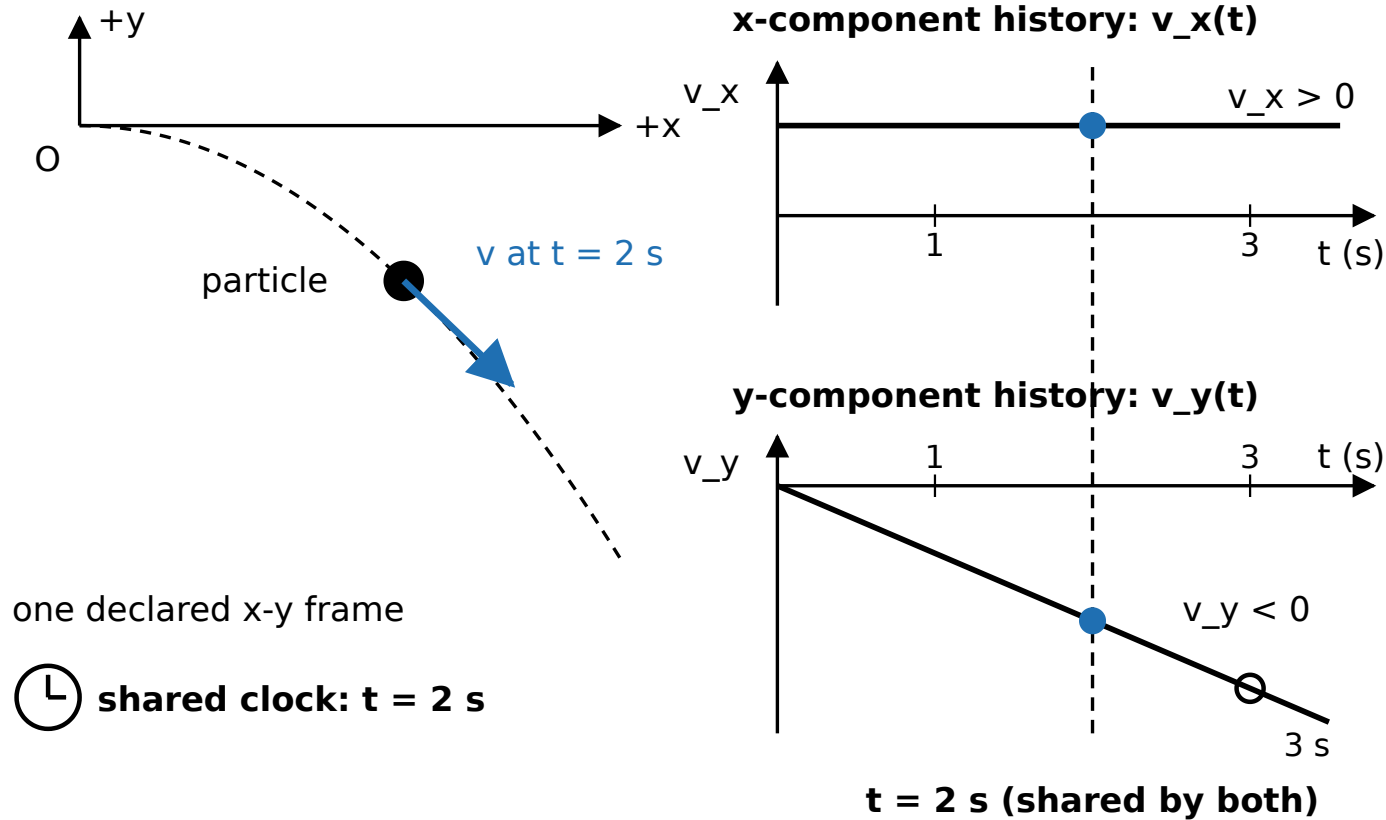
AUTHORED practice

Motion in 2D component and projectile practice

A particle starts with velocity $(3 \mathbf{i} + 4 \mathbf{j})$ m/s and has constant acceleration $(2 \mathbf{i} - 1 \mathbf{j})$ m/s² for 3 s. Find its velocity and displacement after 3 s.

Conditions

- Both acceleration components are constant over the 3 s interval.
- The initial position is irrelevant because the question asks for displacement.



✓ $(v_x(2\text{ s}), v_y(2\text{ s}))$: simultaneous, one valid plane state

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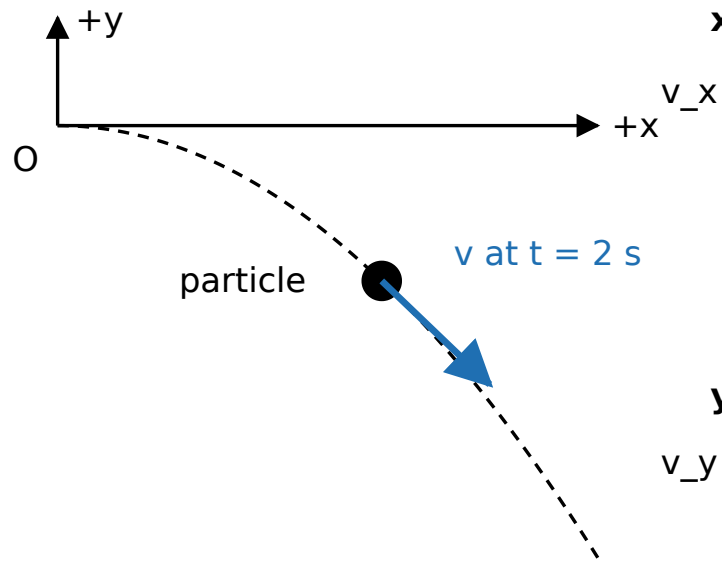
Keep independently evolving x and y descriptions attached to one physical object, one declared frame and one shared clock, so only simultaneous component values are recombined into a plane state.

1. Write the x and y initial velocity and acceleration values separately before calculating.

Reasoning route and full solution

1. **REPRESENT** Resolve the vectors into $u_x=3$, $u_y=4$, $a_x=2$ and $a_y=-1$, with the same elapsed time 3 s on both axes.
Why valid: Component kinematics is valid only after the signed components and common physical interval are explicit.
2. **CONNECT** Apply $v=u+at$ and $\text{displacement}=ut+(1/2)at^2$ separately to x and y while preserving signs and the common time.
Why valid: The given acceleration vector is constant, so each Cartesian acceleration component is constant over the same interval and the 1D relations apply component-wise.
3. **TRANSFORM** Compute $v_x=9$, $v_y=1$, $\Delta x=18$ and $\Delta y=7.5$ from the two component equations.
Why valid: Substitution into the component equations preserves the negative y acceleration instead of replacing signed components by magnitudes.
4. **VERIFY** Check units and signs axis by axis, then confirm both vector components refer to the same 3 s endpoint.

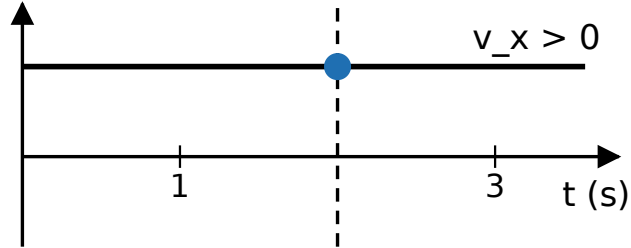
Why valid: Unit consistency detects relation misuse and the common-endpoint check prevents combining component states from different times.



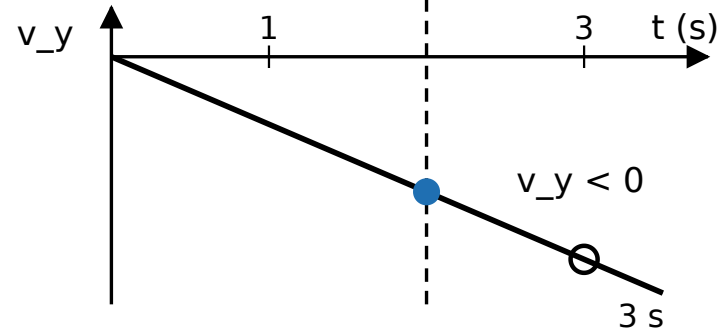
one declared x-y frame

⌚ shared clock: $t = 2$ s

x-component history: $v_x(t)$



y-component history: $v_y(t)$



$t = 2$ s (shared by both)

✓ $(v_x(2 \text{ s}), v_y(2 \text{ s}))$: simultaneous, one valid plane state

✗ $(v_x(2 \text{ s}), v_y(3 \text{ s}))$: non-simultaneous, rejected (no single instant)

Keep independently evolving x and y descriptions attached to one physical object, one declared frame and one shared clock, so only simultaneous component values are recombined into a plane state.

1. Use $v_x = 3 + 2(3) = 9$ and $v_y = 4 - 1(3) = 1$.
2. Use $\Delta x = 3(3) + 0.5(2)(3^2) = 18$.
3. Use $\Delta y = 4(3) + 0.5(-1)(3^2) = 7.5$.
4. Both axes use the same 3 s interval.

Answer

$v = (9 \text{ i} + 1 \text{ j}) \text{ m/s}$ and $\Delta r = (18 \text{ i} + 7.5 \text{ j}) \text{ m}$.

Independent check

Each axis uses only its own signed component values; units are m/s for velocity and m for displacement.

If you went wrong

You got $v = (9 \text{ i} + 7 \text{ j}) \text{ m/s}$ or $\Delta y = 16.5 \text{ m}$: the -1 m/s^2 was used as $+1$, so the y-acceleration lost its sign.

Repair

Revisit: Apply the one-dimensional constant-acceleration relations independently to x and y using only that axis's signed values and the same t .

What this establishes

Established: with both acceleration components constant, the one-dimensional relations run once per axis on a shared interval with their signs kept. The transfer task asks you first to decide, axis by axis, whether that condition holds at all.

AUTHORED practice

Motion in 2D component and projectile practice

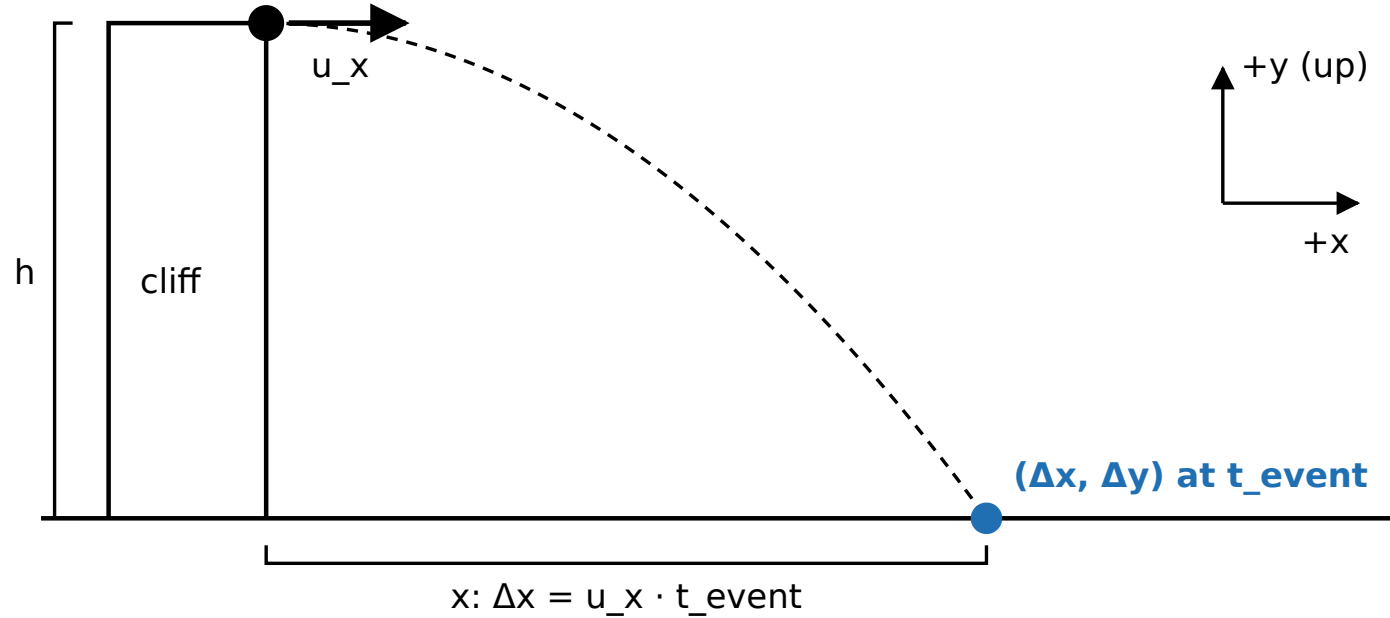
An ideal projectile is launched with already-resolved components $u_x=12$ m/s and $u_y=20$ m/s. Take +y upward and $g=10$ m/s². Find the time to the apex, the velocity at the apex, the acceleration at the apex, and the height gained above launch.

Conditions

- Air resistance is neglected.
- After release, gravity is the only retained force.
- The given initial components are already resolved.

Requested event: the stone reaches the ground

event condition (y): $\Delta y = -h$



one shared event timeline

event time t_{event}



y: $\Delta y = -h$ at the same t_{event}

The one t_{event} is reused unchanged on x and on y.

Make event selection explicit: one component condition identifies the requested event time, and that same time is then reused on every component before reconstructing the requested state.

1. Which component condition defines the apex?

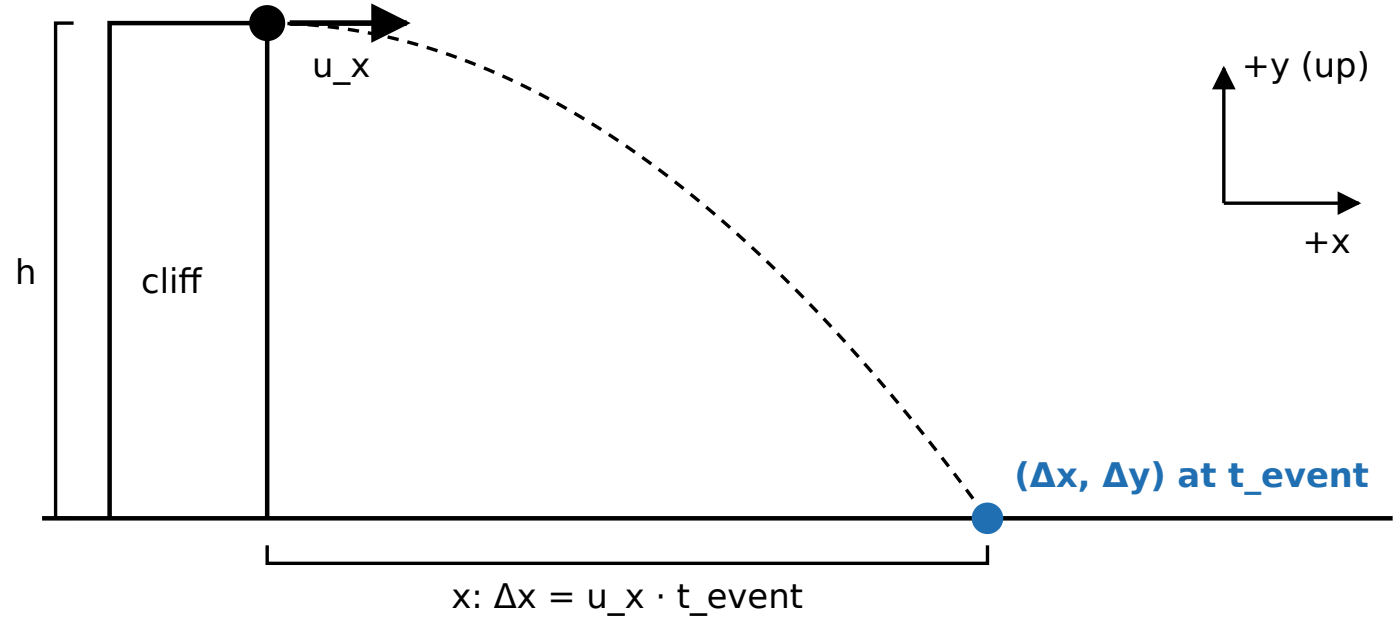
Reasoning route and full solution

1. **REPRESENT** Represent ideal flight with $a_x=0$, $a_y=-g$, constant $v_x=u_x$ and $v_y=u_y-gt$ in the upward-positive frame.
Why valid: Gravity-only free flight leaves horizontal velocity constant while vertical velocity changes under a constant downward acceleration.
2. **DECIDE** Identify the apex by $v_y=0$, not by setting the full velocity or acceleration to zero.
Why valid: At a vertical turning point v_y changes sign, while horizontal motion continues and gravity remains present.
3. **TRANSFORM** Solve $t=2$ s, then evaluate velocity, acceleration and height gain at that same event time.
Why valid: Every requested quantity describes the same apex event, so the time obtained from $v_y=0$ must be reused across both components.
4. **VERIFY** Confirm that only v_y vanishes at the apex while v_x and downward acceleration remain nonzero.

Why valid: This component check distinguishes a vertical turning point from the false claim that the projectile stops or gravity vanishes.

Requested event: the stone reaches the ground

event condition (y): $\Delta y = -h$



one shared event timeline

event time t_{event}



y: $\Delta y = -h$ at the same t_{event}

The one t_{event} is reused unchanged on x and on y.

Stage 1 of 4

Make event selection explicit: one component condition identifies the requested event time, and that same time is then reused on every component before reconstructing the requested state.

1. At the apex set $v_y = u_y - gt = 0$, so $t = 20/10 = 2$ s.
2. Horizontal acceleration is zero, so v_x remains 12 m/s.
3. Gravity still acts at the apex, so $a_y = -10$ m/s².
4. $\Delta y = 20(2) - 0.5(10)(2^2) = 20$ m.

Answer

The apex occurs at $t = 2$ s. There $v = (12 \mathbf{i} + 0 \mathbf{j})$ m/s, $a = (0 \mathbf{i} - 10 \mathbf{j})$ m/s², and the height gained is 20 m.

Independent check

Only v_y is zero at the apex; neither the full velocity nor the acceleration is zero.

If you went wrong

Your apex line says the velocity or the acceleration is zero: only v_y vanishes there, so a zero speed at the top means the 12 m/s horizontal component was dropped, and a zero acceleration means gravity was switched off.

Repair

Revisit: For an apex event, set $v_y=0$ while keeping $v_x=u_x$ and $a_y=-g$.

What this establishes

Established: the apex is the event $v_y = 0$, with v_x unchanged and $a_y = -g$ still acting. The transfer task compares two equal-height events on one path using the same component facts, without solving for any time.

AUTHORED practice

Motion in 2D component and projectile practice

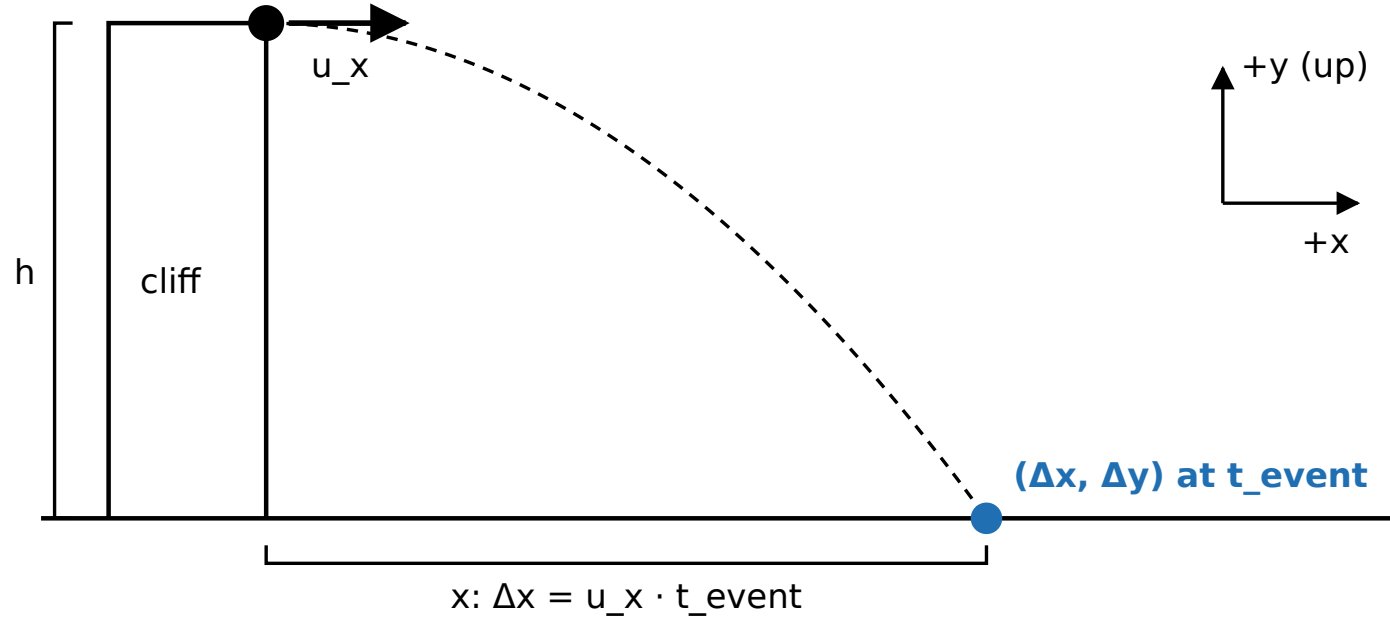
A stone is launched horizontally at 15 m/s from a cliff 20 m above level ground. Take +y upward and $g=10 \text{ m/s}^2$. Find the flight time, horizontal range and impact velocity.

Conditions

- Air resistance is neglected.
- The launch has $u_y=0$.
- Ground is 20 m below the launch point.

Requested event: the stone reaches the ground

event condition (y): $\Delta y = -h$



one shared event timeline

event time t_{event}



y: $\Delta y = -h$ at the same t_{event}

The one t_{event} is reused unchanged on x and on y.

Make event selection explicit: one component condition identifies the requested event time, and that same time is then reused on every component before reconstructing the requested state.

1. Which component equation contains the known 20 m drop?

Reasoning route and full solution

1. **REPRESENT** Represent the horizontal and vertical histories in one declared frame with one shared elapsed time, placing the 20 m drop in the vertical component.

Why valid: The component equations evolve independently but still describe one stone at one physical instant, so the known landing geometry belongs to y while time remains common.

2. **DECIDE** Use the vertical ground-contact condition $\Delta y = -20$ m to determine the flight time before calculating horizontal range or impact velocity.

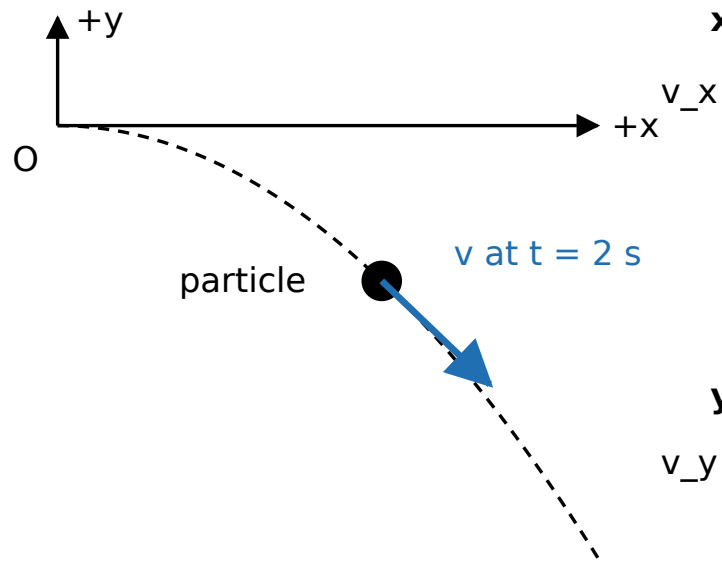
Why valid: Ground impact is defined by the vertical displacement, so the y equation identifies the event time; that one time then belongs to every component at impact.

3. **CONNECT** Reuse $t_{\text{event}} = 2$ s in the horizontal displacement and both velocity components.

Why valid: Range and impact velocity must be reconstructed at the same physical event that the vertical equation identified.

4. **VERIFY** Check that the ideal fall time is set by vertical motion while horizontal speed changes the range but not the vertical landing event.

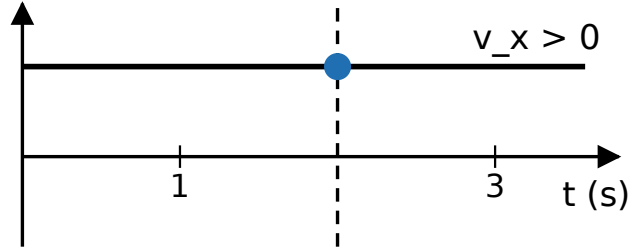
Why valid: With $a_x = 0$ and no drag, the vertical event equation contains no horizontal speed.



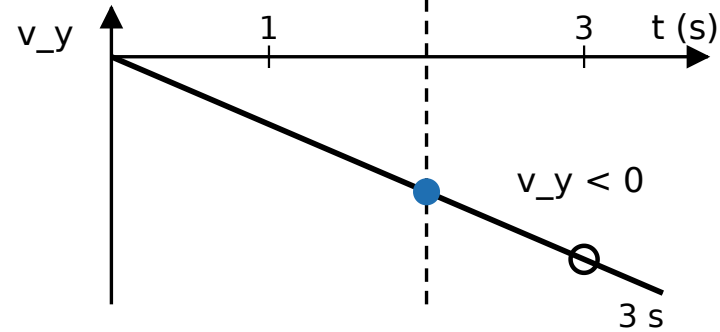
one declared x-y frame

⌚ shared clock: $t = 2$ s

x-component history: $v_x(t)$



y-component history: $v_y(t)$



$t = 2$ s (shared by both)

✓ $(v_x(2 \text{ s}), v_y(2 \text{ s}))$: simultaneous, one valid plane state

✗ $(v_x(2 \text{ s}), v_y(3 \text{ s}))$: non-simultaneous, rejected (no single instant)

Keep independently evolving x and y descriptions attached to one physical object, one declared frame and one shared clock, so only simultaneous component values are recombined into a plane state.

1. Use the vertical condition $\Delta y = -20 = 0 \cdot t - 0.5(10)t^2$, giving $t = 2$ s.
2. Then $\Delta x = 15(2) = 30$ m.
3. $v_x = 15$ m/s and $v_y = -10(2) = -20$ m/s at impact.

Answer

The flight time is 2 s, the horizontal range is 30 m, and the impact velocity is $(15 \mathbf{i} - 20 \mathbf{j})$ m/s.

Independent check

The fall time comes from vertical motion; changing only horizontal speed would change range but not the ideal fall time.

If you went wrong

Your time came from the 15 m/s horizontal speed, for example $20/15 \approx 1.33$ s giving a 20 m range: the fall time must come from the 20 m drop alone.

Repair

Revisit: For any unequal-height landing, use the actual $\Delta y = y_f - y_i$ in the vertical position equation and retain the actual u_y ; for a horizontal launch specifically, set $u_y = 0$, then obtain the event time

[vertically and share it with x.](#)

What this establishes

Established: the drop fixes the fall time through the vertical equation, and that one time then sets the range. The transfer task asks whether the projectile model itself still holds when a force keeps acting after release.

AUTHORED practice

Motion in 2D component and projectile practice

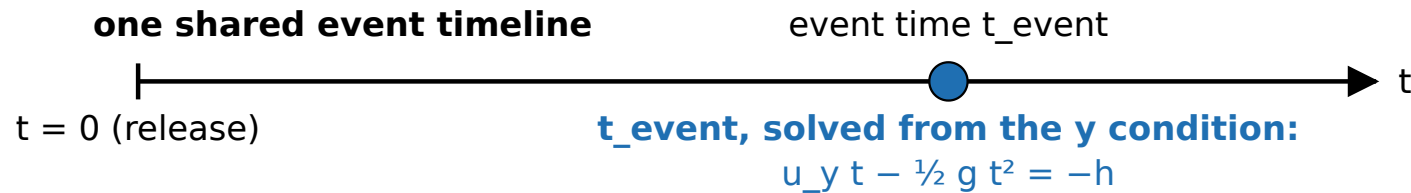
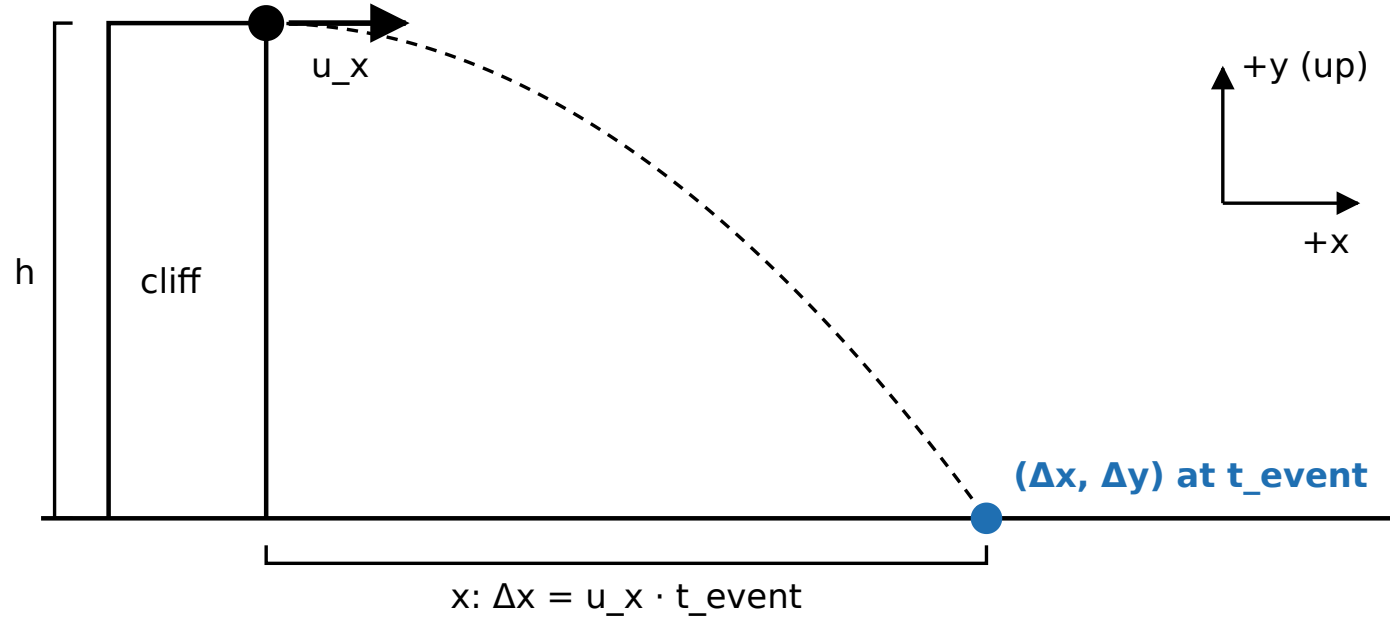
An ideal projectile is launched and later lands at the same height with $u_x=12$ m/s, $u_y=16$ m/s and $g=8$ m/s². Find the total flight time, range and landing velocity.

Conditions

- Air resistance is neglected.
- Launch and landing heights are equal.
- Take +y upward.

Requested event: the stone reaches the ground

event condition (y): $\Delta y = -h$



y: $\Delta y = -h$ at the same t_{event}

The one t_{event} is reused unchanged on x and on y.

Make event selection explicit: one component condition identifies the requested event time, and that same time is then reused on every component before reconstructing the requested state.

1. What vertical displacement condition is true at a same-height landing?

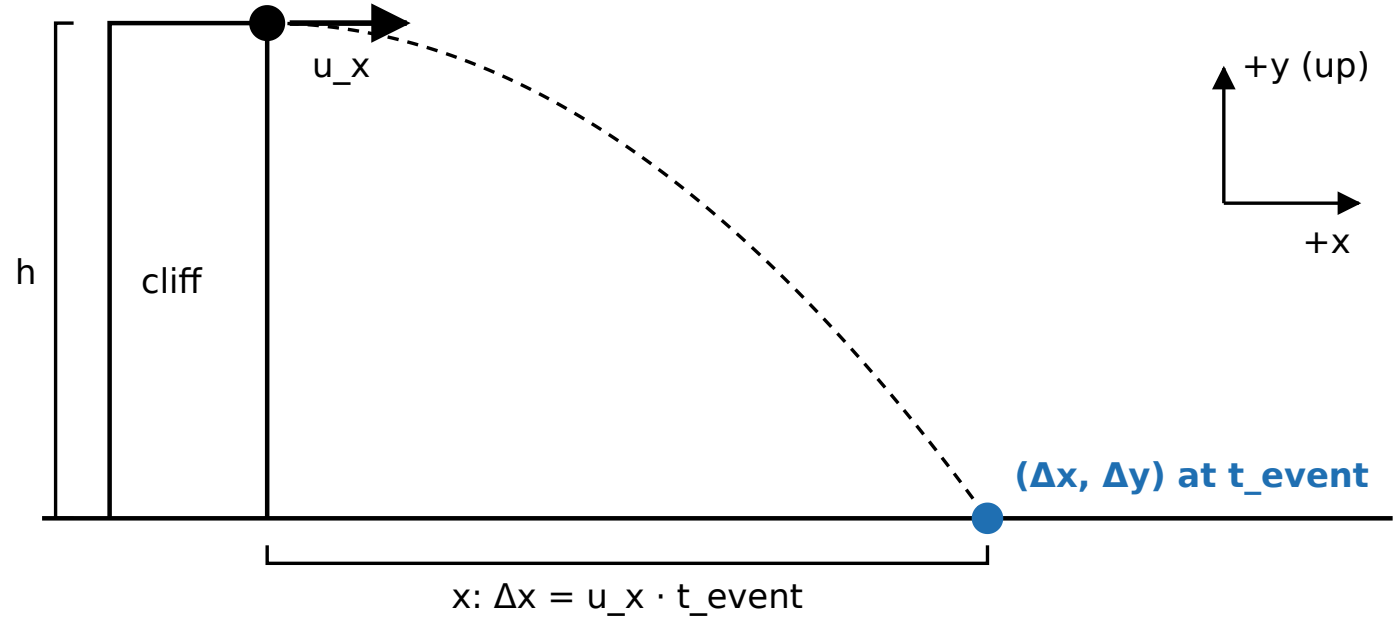
Reasoning route and full solution

1. **REPRESENT** Represent landing at the launch height with $\Delta y=0$ while keeping $a_x=0$, $a_y=-g$ and one common flight time.
Why valid: Same launch and landing height fixes the vertical event displacement without changing the gravity-only component model.
2. **DECIDE** Choose the nonzero root of the vertical event equation as the landing time and reject $t=0$ as the launch event.
Why valid: The equation describes two visits to the launch height; event identity selects the later root because the question asks for landing.
3. **CONNECT** Reuse $t=4$ s in horizontal displacement and vertical velocity to obtain range and landing velocity.
Why valid: Range and landing velocity belong to the same landing event identified by the vertical equation.
4. **VERIFY** Compare launch and landing components: v_x must match while v_y has equal magnitude and opposite sign.

Why valid: With zero horizontal acceleration and constant downward gravity, return to the same height restores vertical speed magnitude while reversing its direction.

Requested event: the stone reaches the ground

event condition (y): $\Delta y = -h$



one shared event timeline

event time t_{event}



y: $\Delta y = -h$ at the same t_{event}

The one t_{event} is reused unchanged on x and on y.

Stage 1 of 4

Make event selection explicit: one component condition identifies the requested event time, and that same time is then reused on every component before reconstructing the requested state.

1. At landing $\Delta y = 0 = 16t - 0.5(8)t^2$; the nonzero root is $t = 4$ s.
2. Horizontal velocity stays 12 m/s, so range = $12(4) = 48$ m.
3. $v_y = 16 - 8(4) = -16$ m/s at landing.

Answer

The flight time is 4 s, the range is 48 m, and the landing velocity is $(12 \mathbf{i} - 16 \mathbf{j})$ m/s.

Independent check

At the same height, the horizontal component matches launch and the vertical component has equal magnitude with opposite sign.

If you went wrong

Your flight time is 2 s and your range 24 m: 2 s is u_y/g , the time to the apex, which is half the flight; $\Delta y = 0$ has its non-zero root at 4 s.

Repair

Revisit: For a return to the launch height, set $\Delta y = 0$ and treat any same-height flight-time/range shortcut as conditional on that equality.

What this establishes

Established: a same-height landing is the non-zero root of $\Delta y = 0$, and the range uses that same time. The transfer task drops the equal-height condition, so the landing event has to be written from the actual drop.